投影定理

$Prj_u(a+b) = Prj_u a + Prj_u b$

证明 柯西-施瓦茨不等式 设 $a_i \in R$, $b_i \in R$ $(i=1,2,3,\dots n)$ 有 (离散版)

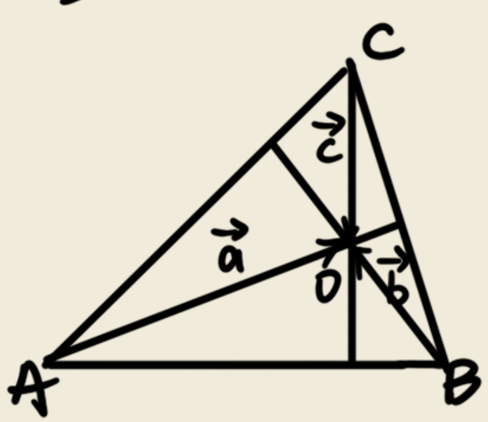
$$\left(\sum_{i=1}^{n} a_i b_i\right)^2 \le \left(\sum_{i=1}^{n} a_i^2\right) \cdot \left(\sum_{i=1}^{n} b_i^2\right)$$

令 $\vec{a} = (a_1, a_2, \dots, a_n)$ $\vec{b} = (b_1, b_2, \dots, b_n)$

左 $= (\vec{a} \cdot \vec{b})^2 \le |\vec{a}|^2 \cdot |\vec{b}|^2 =$ 右

$$\left(\int_a^b f(x) \cdot g(x)\, dx\right)^2 \le \left(\int_a^b f^2(x)\, dx\right) \cdot \left(\int_a^b g^2(x)\, dx\right)$$ 连续版

用向量证明三条高线交于一点



$\vec{a} \cdot \overrightarrow{BC} = 0 \Rightarrow \vec{a} \cdot (\vec{b} - \vec{c}) = 0$ ②

$\vec{b} \cdot \overrightarrow{AC} = 0 \quad\ \ \vec{b} \cdot (\vec{a} - \vec{c}) = 0$ ①

$\Rightarrow \vec{c} \cdot \overrightarrow{AB} = \vec{c} \cdot (\vec{a} - \vec{b}) =$ ① $-$ ② $= 0$

$\vec{i}\times\vec{i}=\vec{j}\times\vec{j}=\vec{k}\times\vec{k}=\vec{0}$

$a=(a_1,a_2,a_3)$, $b=(b_1,b_2,b_3)$

$$a\times b=\left(\begin{vmatrix}a_2 & a_3\\ b_2 & b_3\end{vmatrix},\begin{vmatrix}a_3, & a_1\\ b_3, & b_1\end{vmatrix},\begin{vmatrix}a_1, & a_2\\ b_1, & b_2\end{vmatrix}\right)=\begin{vmatrix}i & j & k\\ a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\end{vmatrix}=\begin{vmatrix}a_2 & a_3\\ b_2 & b_3\end{vmatrix}\vec{i}-\begin{vmatrix}a_1 & a_3\\ b_1 & b_3\end{vmatrix}\vec{j}+\begin{vmatrix}a_1 & a_2\\ b_1 & b_2\end{vmatrix}\vec{k}$$

$$=(a_1b_2-a_2b_1)\underbrace{i\times j}_{k}+(a_2b_3-a_3b_2)\underbrace{j\times k}_{i}+(a_3b_1-a_1b_3)\underbrace{k\times i}_{j}$$

证明海仑公式

$$S=\sqrt{p(p-a)(p-b)(p-c)}$$

$$S^2=\frac{1}{4}|\vec{a}\times\vec{b}|^2=\frac{1}{4}|\vec{a}|^2|\vec{b}|^2\sin^2\langle\vec{a},\vec{b}\rangle=\frac{1}{4}|\vec{a}|^2|\vec{b}|^2(1-\cos^2\langle\vec{a},\vec{b}\rangle)$$

而 $\cos\langle\vec{a},\vec{b}\rangle=\frac{a^2+b^2-c^2}{2ab}$

$$=\frac{1}{4}a^2b^2\left(1-\left(\frac{a^2+b^2-c^2}{2ab}\right)^2\right)=\frac{1}{16}\left(4a^2b^2-(a^2+b^2-c^2)^2\right)=\frac{1}{16}\left((a+b)^2-c^2\right)\left(c^2-(a-b)^2\right)=p(p-a)(p-b)(p-c)$$

向量的混合积

$(a,b,c)=a\times b\cdot c$　以 a,b 为底的有向平行六面体的体积

性质 $(a,b,c)=(b,c,a)=(c,a,b)=-(a,c,b)=-(b,a,c)=-(c,b,a)$

$(a,b,\lambda c+\mu d)=\lambda(a,b,c)+\mu(a,b,d)$

$(a,b,c)=0$ 的充要条件是向量 a,b,c 共面

证明 $(a\times b)\times c=(a\cdot c)b-(b\cdot c)a$

设 $a=(a_1,a_2,a_3)$, $b=(b_1,b_2,b_3)$, $c=(c_1,c_2,c_3)$

$(a\times b)\times c=(x_1,x_2,x_3)$

$$a\times b=\left(\begin{vmatrix}a_2 & a_3\\ b_2 & b_3\end{vmatrix},\begin{vmatrix}a_3 & a_1\\ b_3 & b_1\end{vmatrix},\begin{vmatrix}a_1 & a_2\\ b_1 & b_2\end{vmatrix}\right)$$

$$x_1=\begin{vmatrix}\begin{vmatrix}a_3 & a_1\\ b_3 & b_1\end{vmatrix} & \begin{vmatrix}a_1 & a_2\\ b_1 & b_2\end{vmatrix}\\ c_2 & c_3\end{vmatrix}=c_3(a_3b_1-a_1b_3)-c_2(a_1b_2-a_2b_1)$$

$$=(a_1c_1+a_2c_2+a_3c_3)b_1-(b_1c_1+b_2c_2+b_3c_3)a_1$$

$$=(a\cdot c)b_1-(b\cdot c)a_1$$

同理 $x_2=(a\cdot c)b_2-(b\cdot c)a_2$

$x_3=(a\cdot c)b_3-(b\cdot c)a_3$